SAFTER: Safety-Aware Filtering Through Explicit Recoverability for Robust Reinforcement Learning under Distribution Shift

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Abstract: Learning safe policies that remain reliable in unseen environments is challenging under bounded actuation, where satisfying current-state constraints may still leave insufficient control authority for future recovery. We propose SAFTER (**S**afety-**A**ware **F**iltering **T**hrough **E**xplicit **R**ecoverability), a framework that embeds a differentiable closed-form backup-CBF filter into Soft Actor-Critic and combines current safety, sampled backup safety, terminal recovery, and input admissibility into a single differentiable certificate. The resulting affine condition admits an exact minimum-intervention projection in surrogate-control space without iterative online optimization, enabling direct policy learning through the safety filter. In quadruped navigation, SAFTER reaches the same training-performance criterion with 30.6% fewer transitions than Vanilla SAC and 34.1–35.1% fewer than current-state CBF-filtered SAC baselines. Under geometric distribution shift, SAFTER achieves 88.6% success in simulation, exceeding the strongest baseline by 38.0 percentage points, and 73.3% and 76.7% success on a physical quadruped in two matched laboratory layouts, improving over the strongest baselines by 10.0 and 60.0 points, respectively. Moreover, ablations show that differentiating through the exact safety-filter Jacobian is the key learning mechanism behind SAFTER's robustness, increasing success from 6.2% to 61.4% in the most challenging shifted environment. Videos are available on the project website.

Keywords: Safe Reinforcement Learning, Backup Control Barrier Functions, Recoverability-Aware Safety Filtering

1 Introduction

Reinforcement learning (RL) enables adaptive navigation directly from local sensing [1, 2], but trial-and-error interaction can generate unsafe actions during both training and deployment [3, 4]. Reward shaping can encourage safer behavior [5, 6, 7], but does not explicitly prevent unsafe actions from being executed [8]. This limitation motivates explicit safety mechanisms that constrain or filter policy actions during learning and deployment [9, 10, 11, 12, 13, 14]. Control Barrier Functions (CBFs) provide one such mechanism by minimally modifying nominal actions to satisfy prescribed safety constraints [15], and have been incorporated into RL through analytical filters, sensing-based navigation, and barrier-inspired learning objectives [16, 17, 18, 19]. However, conventional CBFs primarily enforce current-state safety: under bounded actuation, an action can be safe now while driving the system toward a state from which future recovery is no longer feasible. Backup CBFs (bCBFs) address this limitation by requiring a prescribed backup maneuver to remain safe over a finite horizon and reach a backup-invariant terminal set [20, 21, 22]. Time-varying and learning-based backup strategies can reduce conservatism [23, 24, 25], while closed-form nominal–backup interpolation avoids online QP solving [26]. Reachability-based methods similarly reason about

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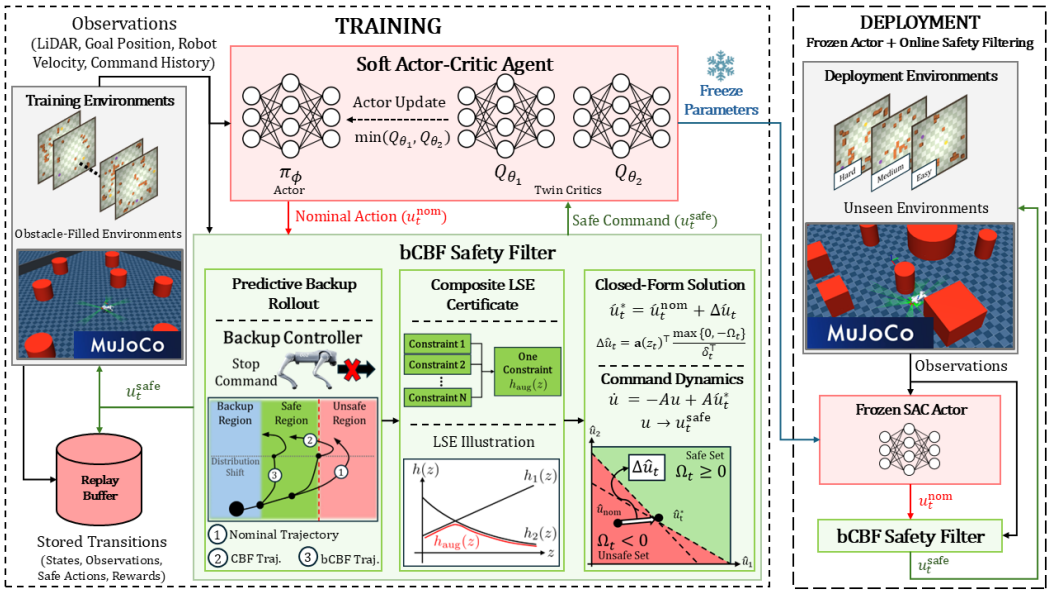


Figure 1: Overview of SAFTER. Backup prediction and hierarchical constraint composition yield an analytical minimum-intervention safety filter. The critic evaluates executed actions, while the actor differentiates through the safety map during training.

future recoverability [27, 28]. Nevertheless, predictive safety introduces many future, terminal, and input constraints, creating a computational and differentiation bottleneck when the filter must be evaluated repeatedly inside RL training.

To address these coupled challenges, we propose **SAFTER**, which preserves finite-horizon recoverability under bounded commands, reduces the resulting safety and recovery constraints to an explicit minimum-intervention backup-CBF projection, and retains the exact filter Jacobian for policy learning. This construction avoids iterative online optimization while enabling SAC to learn through the recoverability-aware filter, improving sample efficiency and robustness under geometric distribution shift in simulation and physical quadruped experiments.

2 Recoverability-Aware Differentiable Safety Filtering

Setting and design objective. We consider $\dot{x} = f(x) + g(x)u$, where $x \in \mathbb{R}^n$ is the system state, $u \in \mathbb{R}^m$ is the command, and f and g denote the drift and input vector fields. Commands satisfy $\mathcal{U} = \{u : u_{\min} \leq u \leq u_{\max}\}$, with fixed componentwise bounds. At decision step t, a Soft Actor-Critic (SAC) policy [29], parameterized by ϕ, proposes $u_t^{\text{nom}} \sim \pi_\phi(\cdot \mid o_t)$ from observation o_t in environments that may differ from the training distribution. Safety is specified by N_h differentiable functions h_i through $\mathcal{C} = \{x : h_i(x) \geq 0,\ i = 1, \ldots, N_h\}$. Our objective is to design a sensing-based safety filter that minimally modifies u_t^{nom} to keep the system within $\mathcal{C}$ under bounded commands and distribution shift, while remaining efficient to evaluate and differentiate during policy learning. Figure 1 summarizes the proposed SAFTER architecture.

Command augmentation and backup recoverability. To reason about safety under bounded commands, we augment the applied command u with first-order dynamics $\dot{u} = -Au + A\hat{u}$, where $\hat{u} \in \mathbb{R}^m$ is the surrogate input and $A \succ 0$ is diagonal [30]. With $z = [x^\top, u^\top]^\top$, the augmented dynamics are $\dot{z} = \hat{f}(z) + \hat{g}(z)\hat{u}$, where $\hat{f}(z) = [(f(x) + g(x)u)^\top, (-Au)^\top]^\top$ and $\hat{g}(z) = [0_{n\times m}^\top, A^\top]^\top$. The physical bounds $u \in \mathcal{U}$ are enforced through the augmented command dynamics. Since h_i has relative degree two with respect to $\hat{u}$, current safety is represented by $\psi_i(z) = L_f h_i(x) + L_g h_i(x)u + k_0 h_i(x)$, with $k_0 > 0$. Current safety alone does not ensure future recoverability. Let $\phi_b^z(\tau, z)$ denote the augmented backup flow under $\dot{z} = \hat{f}(z) + \hat{g}(z)\hat{u}_b$, where $\hat{u}_b$

is the prescribed backup input, $\tau \in [0, T]$ is prediction time, and $T > 0$ is the backup horizon. Let $\mathcal{C}_B^z$ denote the backup-invariant terminal set and $\phi_b^x(\tau, z)$ the original-state component of the backup flow. For sampled times $0 < \tau_1 < \cdots < \tau_K = T$, recoverability requires $\phi_b^x(\tau_k, z) \in \mathcal{C}$ for all k and $\phi_b^z(T, z) \in \mathcal{C}_B^z$ [20, 21]. Appendix B.1 provides additional details.

Composite certificate. The resulting conditions include current safety, sampled backup safety, terminal recovery, and command admissibility. For a collection S of scalar constraint functions and $\kappa > 0$, define the Log-Sum-Exp (LSE) smooth minimum as $s_\kappa(S) = -\kappa^{-1} \log \sum_{c \in S} e^{-\kappa c}$ [31]. Hierarchical LSE composition combines these conditions into a single differentiable certificate $h_{\text{aug}}(z)$, reducing their enforcement to one affine CBF constraint. Appendix B.2 gives the complete construction.

Analytical safety projection. Define $a(z) = L_{\hat{g}} h_{\text{aug}}(z)$ and $b(z) = L_{\hat{f}} h_{\text{aug}}(z) + \alpha_{\text{aug}} h_{\text{aug}}(z)$, where $\alpha_{\text{aug}} > 0$, so that $a(z)\hat{u} + b(z) \geq 0$. The actor command u_t^{nom} is mapped to a nominal surrogate $\hat{u}_t^{\text{nom}}$ as detailed in Appendix D. At step t, the physical command bounds induce an admissible surrogate set $\hat{\mathcal{U}}_t$, giving

$$\hat{u}_t^\star = \arg\min_{\hat{u} \in \hat{\mathcal{U}}_t} \frac{1}{2} \|\hat{u} - \hat{u}_t^{\text{nom}}\|_2^2, \quad \text{s.t.} \quad a(z_t)\hat{u} + b(z_t) \geq 0. \tag{1}$$

When the surrogate bounds are inactive, its closed-form solution is

$$\hat{u}_t^\star = \hat{u}_t^{\text{nom}} + a(z_t)^\top \frac{\max\{0, -\Omega_t\}}{a(z_t)a(z_t)^\top}, \qquad \Omega_t = a(z_t)\hat{u}_t^{\text{nom}} + b(z_t), \tag{2}$$

for $a(z_t)a(z_t)^\top > 0$. Bound-active cases retain the same analytical minimum-intervention structure over the corresponding box face. Derivations, implementation, and guarantee scope are provided in Appendices B.3, B.4, and D.

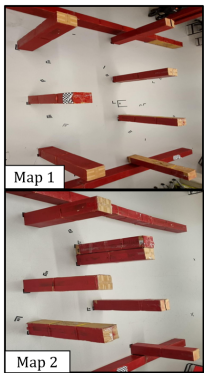
Theorem 2.1 (Continuous-time safety and sampled recoverability)**.** *Assume continuously differentiable augmented dynamics and certificate, a backup-invariant terminal set $\mathcal{C}_B^z$, and a well-posed closed loop continuously enforcing $a(z)\hat{u} + b(z) \geq 0$. If $x(0) \in \mathcal{C}$ and $h_{\text{aug}}(z(0)) \geq 0$, then $h_{\text{aug}}(z(t)) \geq 0$, $x(t) \in \mathcal{C}$, and $u(t) \in \mathcal{U}$ throughout the solution's interval of existence. The predicted backup trajectory also satisfies the selected horizon constraints and reaches $\mathcal{C}_B^z$ at T.*

Differentiable SAC integration. The actor command u_t^{nom} is mapped to $\hat{u}_t^{\text{nom}}$, filtered to $\hat{u}_t^\star$ through (1), and propagated through the command dynamics to obtain the executed command u_t^{safe}. The replay buffer stores the executed action and resulting transition, so the critics learn from the action that actually affects the environment. During actor updates, gradients propagate through the same analytical filter, exposing the exact local effect of the recoverability-aware intervention to the policy. Appendix E provides the complete SAC formulation.

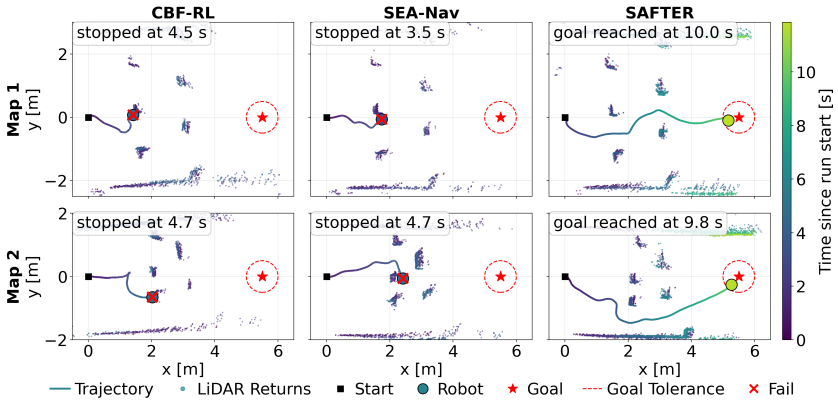
3 Experimental Results

Training Sample Efficiency. The training were done using Unitree Go2 in MuJoCo 3.3.6 with 256 parallel environments and five independent seeds, each budgeted 20.48M transitions. The predictive backup recoverability implemented in SAFTER improves sample efficiency, defined by the first attainment of 90% training success, where it reaches the threshold in 4.06 ± 0.08M transitions, 30.6% and 34.1%–35.1% fewer than Vanilla SAC and current-state CBF baselines, respectively. Further details on maps, training, runtime, rewards, and SAC implementation are provided in Appendices A.1–E.4.

Robustness under Geometric Distribution Shift. We evaluate the trained policies on two real-world maps shown in Figure 2a and five training and three unseen simulation maps. The trajectories in Figure 2b illustrate the effect of backup recoverability. SAFTER takes earlier, wider detours around obstacle clusters and preserves clearance to the goal, whereas CBF–RL and SEA–Nav enter tighter regions and fail. This behavior is consistent with SAFTER rejecting actions that would leave insufficient room for a safe backup maneuver, thereby maintaining a recoverable path under the shifted geometry. Simulation and real-robot results summarized in Table 1(a) further show



(a) Real-world evaluation maps.



(b) Trajectory sample and LiDAR readings of the real robot experiments.

Figure 2: Real-world map with two different obstacle configuration, and trajectory sample and LiDAR readings taken from the Unitree Go2 on-board state estimator and LiDAR.

Table 1: Navigation success rates (%). Simulation values are mean ± standard deviation across 25 evaluation seeds, while real-world values use 30 trials per map. The † denotes disabled correction rewards that are used to penalize filter intervention.

(a) Robustness in Simulation and Real-robot Tests.

	Simulation		Real robot	
Method	**No-Shift**	**Shifted**	**Map 1**	**Map 2**
SEA–Nav [17]	89.7±2.4	50.6±10.9	56.7	13.3
CBF–RL [18]	84.3±8.2	46.2±11.6	63.3	0.0
Vanilla SAC [29]	74.6±17.8	20.8±6.9	10.0	0.0
CBF–CF [30]	76.6±13.8	48.4±8.4	56.7	16.7
SAFTER (**Ours**)	**95.1±7.1**	**88.6±7.3**	**73.3**	**76.7**

(b) Ablation Study

Environment	**Jacobian**	**Identity**
Low-Shift	**98.8±2.2**	89.8±6.0
Medium-Shift	**88.6±7.3**	41.6±9.3
High-Shift	**61.4±10.2**	6.2±4.4
Low-Shift†	**95.6±4.9**	76.8±7.1
Medium-Shift†	**94.4±4.4**	9.0±6.1
High-Shift†	**76.2±9.2**	2.0±3.5

that SAFTER achieves 88.6% success in the shifted simulation and 73.3% and 76.7% on the two real-robot maps, exceeding the strongest baselines by 38.0, 10.0, and 60.0 percentage points, respectively. Additional experiment setup and results are reported in Appendices A.4–A.6.

Why Does SAFTER Learn Differently? SAFTER shapes the policy through the recoverability-aware safety filter during training, and the exact filter Jacobian determines how this safety correction is propagated back to the actor. Table 1(b) shows that replacing the exact Jacobian with an identity approximation substantially degrades robustness under geometric distribution shift. The effect becomes especially pronounced in the High-Shift environment, where success decreases from 61.4% to 6.2%. The same trend persists without the correction reward, indicating that the exact Jacobian, rather than reward shaping alone, is the dominant mechanism through which SAFTER influences policy learning.

4 Conclusion

SAFTER combines predictive recoverability with analytical safety filtering for policy training. Training results support improved sample efficiency, while the experiments on the real robot and simulations shows the robustness of the policy under geometric distribution shift. The ablation study highlights the ability of the policy to learn the backup controller and safety by enforcing the composite constraint. Future work will address deployment on uneven terrain and dynamic obstacles.

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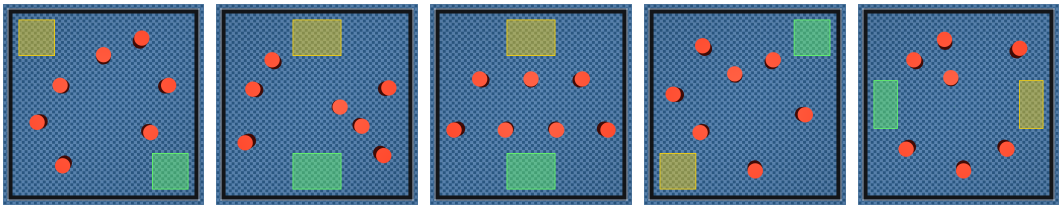
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A Experimental Details and Additional Results

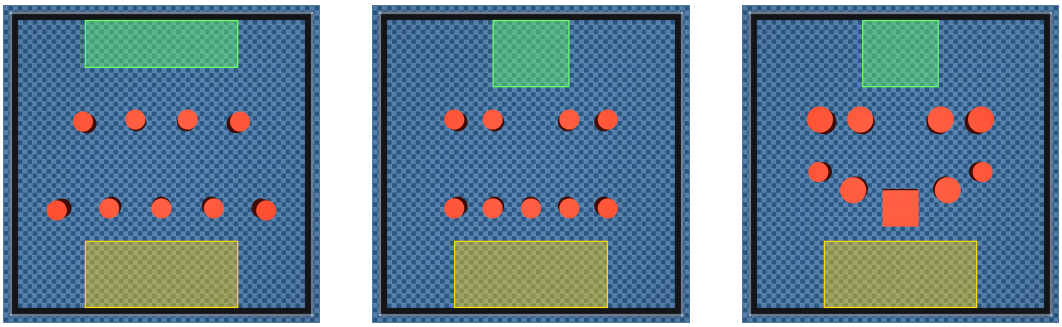
A.1 Training and Evaluation Maps



(a) Training Map 1 (b) Training Map 2 (c) Training Map 3 (d) Training Map 4 (e) Training Map 5

Figure 3: Five hand-crafted training maps with fixed obstacle configurations and predefined initial and goal regions. Green and yellow regions indicate the initial and goal regions, respectively, while obstacles are shown in red.

The quadruped experiments use a fixed $15\ \mathrm{m} \times 15\ \mathrm{m}$ workspace with cylindrical obstacles of radius $0.6\ \mathrm{m}$. The training bank contains the five hand-crafted maps shown in Figure 3, with 200 predefined initial–goal scenarios for each map. These maps define a structured training distribution with varied start–goal orientations and obstacle arrangements. Because the policy receives only a local planar LiDAR observation with a $5\ \mathrm{m}$ maximum range, the complete obstacle configuration is generally not available from the initial state and local navigation decisions are made as new geometry enters the sensing region.



(a) Low-Shift (b) Medium-Shift (c) High-Shift

Figure 4: Static unseen environments used for distribution-shift evaluation. Green and yellow regions indicate the admissible initial and goal regions, respectively, while obstacles are shown in red. The environments introduce increasingly different obstacle configurations relative to the hand-crafted training maps.

The three unseen environments in Figure 4 are constructed to introduce progressively larger geometric shifts relative to the training distribution. The *Low-Shift* environment preserves the general row-based obstacle structure of the training maps while increasing the obstacle count and reversing the initial and goal regions. The *Medium-Shift* environment retains a similar obstacle count but reduces the free-space gaps between obstacle rows, producing narrower navigation passages than those encountered during training. The *High-Shift* environment changes both obstacle geometry and spatial arrangement by introducing different obstacle radii, a square obstacle, and tighter inter-obstacle clearances than the Medium-Shift map.

Table 2: Safety-filter runtime comparison. Total latency includes certificate construction and synthesis, and speedup is measured relative to the direct bCBF–QP. Values are mean ± standard deviation over 300 timed repetitions.

	Batch = 1			Batch = 100		
Method	**Total (ms)**	**Synthesis (ms)**	**Speedup (×)**	**Total (ms)**	**Synthesis (ms)**	**Speedup (×)**
bCBF–QP	30.54±0.32	28.36±0.19	1.00	2504.78±22.46	2487.38±16.94	1.00
bCBF–LSE–QP	5.73±0.45	3.34±0.40	5.33	7.08±0.56	4.58±0.48	353.58
SAFTER	**2.49±0.31**	**0.10±0.01**	**12.2**	**2.52±0.28**	**0.10±0.01**	**990.43**

Table 3: Performance on the training-map distribution with each method's runtime safety filter disabled or enabled. SR, TR, and FR denote success, timeout, and failure rates, respectively. Values are mean ± standard deviation over evaluation seeds and maps.

	Filter disabled			Filter enabled		
Method	**SR ↑**	**TR ↓**	**FR ↓**	**SR ↑**	**TR ↓**	**FR ↓**
SEA–Nav [17]	89.7±2.4	0.0±0.0	10.3±2.4	91.2±2.9	0.0±0.0	8.8±2.9
CBF–RL [18]	84.3±8.2	5.9±5.2	9.8±7.1	84.8±7.4	5.9±5.2	9.3±6.6
CBF–CF [30]	76.6±13.8	2.7±6.3	20.7±15.8	90.1±8.2	1.9±4.5	8.0±8.5
SAFTER (**Ours**)	**95.1±7.1**	0.0±0.0	**4.9±7.1**	**99.7±1.2**	0.0±0.0	**0.3±1.2**

A.2 Additional Training Results

For each seed, we record sample efficiency as the first crossing of 90% success rate the earliest. The mean ± standard deviation of transition counts is 4.06±0.08M for SAFTER, 4.13±0.43M for bCBF–LSE–QP, 5.85±0.47M for Vanilla SAC, 6.16±0.68M for CBF–CF, and 6.26±0.24M for CBF–QP. Figure 5 shows the training success rate across five independent seeds.

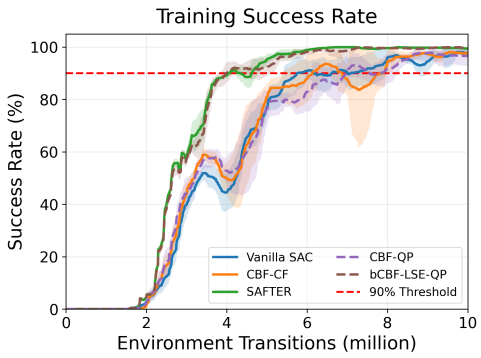


Figure 5: Training success rates of baseline SAC methods and SAFTER under the same training environments. Shaded region denotes the training variance.

A.3 Computation Time Benchmark

To evaluate computational efficiency, a timing comparison was done between SAFTER, bCBF–LSE–QP and the direct unaggregated bCBF–QP. All methods are evaluated on identical fixed filter inputs using 30 warm-up calls and 300 timed repetitions on an NVIDIA RTX 4090 GPU with CUDA synchronization around each timed region. SAFTER achieves the lowest total latency at both batch sizes, providing 12.24× and 990.43× lower latency than the direct bCBF–QP at batch sizes 1 and 100, respectively. Relative to the matched bCBF–LSE–QP, SAFTER is 2.30× and 2.80× faster. Table 2 reports the full timing comparison between the QP-based methods and SAFTER.

A.4 Training Map Policy Testing

Recoverability-aware training also appears to shape the learned nominal policy rather than relying solely on runtime filtering. To examine this effect, we evaluate the trained policies on the training-map distribution with their runtime safety filters enabled and disabled. Among all methods. SAFTER achieves the highest success rate in the training map, with 95.1% success rate without the runtime filter and 99.7% with the filter enabled. The strong filter-disabled performance sug-

Table 4: Navigation performance across unseen static environments. SR, TR, and FR denote success, timeout, and failure rates, respectively. Values are mean ± standard deviation across 25 evaluation-seed rates.

	Low-Shift			Medium-Shift			High-Shift		
Method	**SR ↑**	**TR ↓**	**FR ↓**	**SR ↑**	**TR ↓**	**FR ↓**	**SR ↑**	**TR ↓**	**FR ↓**
SEA–Nav [17]	93.8±6.7	0.0±0.0	6.2±6.7	50.6±10.9	0.0±0.0	49.4±10.9	0.0±0.0	0.0±0.0	100.0±0.0
CBF–RL [18]	61.2±10.7	13.6±6.7	25.2±9.8	46.2±11.6	17.2±7.6	36.6±9.2	31.0±10.0	15.4±8.9	53.6±12.2
Vanilla SAC [29]	34.2±10.1	0.0±0.0	65.8±10.1	20.8±6.9	0.0±0.0	79.2±6.9	17.4±8.3	0.0±0.0	82.6±8.3
CBF–QP [30]	58.8±9.4	0.2±1.0	41.0±9.2	43.2±11.0	0.0±0.0	56.8±11.0	18.0±6.9	0.0±0.0	82.0±6.9
CBF–CF [30]	69.0±11.5	0.0±0.0	31.0±11.5	48.4±8.4	0.0±0.0	51.6±8.4	33.8±12.2	0.0±0.0	66.2±12.2
bCBF–LSE–QP (**Ours**)	**99.4±1.7**	0.0±0.0	**0.6±1.7**	75.0±9.7	0.0±0.0	25.0±9.7	**67.4±11.7**	0.0±0.0	**32.6±11.7**
SAFTER (**Ours**)	98.8±2.2	0.0±0.0	1.2±2.2	**88.6±7.3**	0.0±0.0	**11.4±7.3**	61.4±10.2	0.0±0.0	38.6±10.2

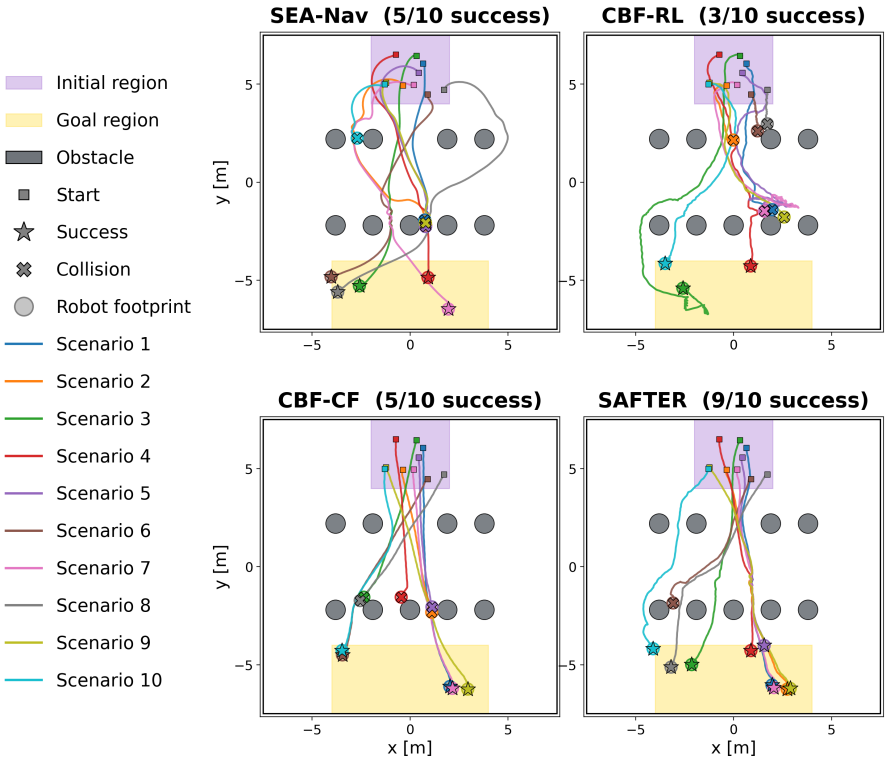


Figure 6: Representative trajectories in the Medium-Shift environment. Purple and yellow regions denote the initial and goal regions, squares mark initial positions, stars successful terminations, crosses collision terminations, and faint circles indicate the robot footprint.

gests that recoverability-aware training promotes nominal behavior compatible with the prescribed backup strategy. The complete performance comparison are reported in Table 3,

A.5 Unseen Maps Evaluation

Each method is evaluated over 500 episodes over 25 seeds. Table 4 extends the main-body comparison to all three shift levels, in which our method achieve higher success rate compared to the baseline methods, while also producing marginal difference compared to the bCBF–LSE–QP, which suggests that the predictive recoverability is the main contributor in maintainin safety under distribution shift. Figure 6 complements these results with example Medium-Shift trajectories for SAFTER, and several baseline methods. Additionally, Table 5 reports path length and navigation time for successful episodes. Compared to the SEA-Nav, SAFTER exhibits longer path length due to a more conser-

Table 5: Navigation efficiency on the unseen static environments with the runtime safety filters enabled. Path Length (PL) and Navigation Time (NT) are computed over successful episodes and reported as mean ± standard deviation. Dashes indicate that no successful episodes were available for computing PL and NT.

	Low-Shift		Medium-Shift		High-Shift	
Method	**PL (m)** ↓	**NT (s)** ↓	**PL (m)** ↓	**NT (s)** ↓	**PL (m)** ↓	**NT (s)** ↓
SEA–Nav [17]	11.57±0.17	13.70±0.17	10.95±0.30	14.21±0.27	−−	−−
CBF–RL [18]	14.86±0.78	32.02±4.02	14.93±0.76	34.93±4.3	15.72±1.19	38.99±8.45
Vanilla SAC [29]	12.76±0.41	26.75±1.54	12.38±0.68	26.31±2.13	12.20±0.70	25.97±1.77
CBF–QP [30]	12.05±0.38	25.08±1.34	11.50±0.45	23.77±1.55	11.82±1.11	25.42±4.02
CBF–CF [30]	12.26±0.36	26.85±1.06	11.72±0.42	24.84±1.22	11.72±0.35	25.93±2.06
bCBF–LSE–QP **(Ours)**	13.06±0.37	32.74±1.11	12.20±0.45	29.50±1.67	12.33±0.54	29.91±2.07
SAFTER **(Ours)**	12.73±0.32	30.66±1.18	11.87±0.38	28.10±1.40	12.03±0.32	29.43±1.34

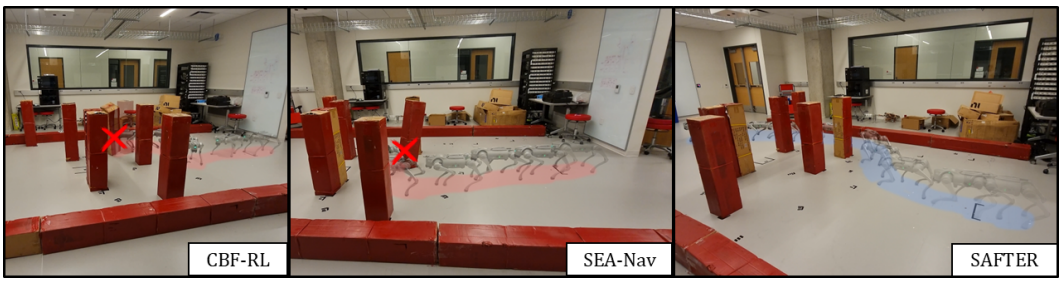


Figure 7: Trajectory comparison of CBF-RL, SEA-Nav and SAFTER in the real map.

vative construction of the barrier functions, with an additional backup margin and velocity-related terms incorporated in the constraints.

A.6 Real-World Experiments

We evaluate the trained navigation policies on a Unitree Go2 in two physical environments containing rectangular obstacles. The layouts are shown in Figure 2a. Within each map, all methods use the same initial position and goal, and each policy is evaluated over 30 trials. Representative trajectories for the selected baseline methods and the proposed SAFTER method are shown in Figure 7. The real experiment result shows how CBF–RL and SEA–Nav struggles to navigate in the real-world environment, while SAFTER is able to incorporate the backup controller and recoverability to avoid colliding with the obstacles placed in the map.

B Theoretical Foundations and Proofs

B.1 Backup Margins and Differentiation

Under the augmented dynamics in Section 2, $L_{\hat{g}} h_i = 0$ and $L_{\hat{g}} \psi_i = L_g h_i A$, so the lifted margin ψ_i has relative degree one wherever $L_g h_i A \neq 0$. Let $\phi_{\mathrm{b}}^z(\tau, z)$ denote the augmented backup flow and $\mathcal{C}_B^z = \{z : h_{B,j}(z) \geq 0,\ j = 1, \ldots, N_B\}$ the terminal set, defined by N_B margins on the augmented state. For $E_x = [I_n\ 0_{n\times m}]$ and $0 < \tau_1 < \cdots < \tau_K = T$, the backup margins are

$$h_i^{\mathrm{b}}(z, \tau_k) = h_i(E_x \phi_{\mathrm{b}}^z(\tau_k, z)), \qquad h_{B,j}^{\mathrm{b}}(z, T) = h_{B,j}(\phi_{\mathrm{b}}^z(T, z)).$$

With backup-flow sensitivity $J_{\mathrm{b}}(\tau, z) = \partial \phi_{\mathrm{b}}^z(\tau, z)/\partial z$, the chain rule gives

$$\nabla_z [c(\phi_{\mathrm{b}}^z(\tau, z))] = J_{\mathrm{b}}(\tau, z)^\top \nabla c(\phi_{\mathrm{b}}^z(\tau, z)),$$

where $c(z) = h_i(E_x z)$ for sampled safety margins or $c(z) = h_{B,j}(z)$ for terminal margins. These gradients enter the composite Lie derivatives. The implementation differentiates the numerical rollout specified in Appendix C.2.

B.2 Hierarchical Certificate and Composite Half-Space

For a finite nonempty indexed collection $\mathcal{S} = \{c_\ell\}_{\ell=1}^M$ and $\kappa > 0$,

$$s_\kappa(\mathcal{S}) = -\frac{1}{\kappa}\log\sum_{\ell=1}^{M} e^{-\kappa c_\ell}, \qquad \min_\ell c_\ell - \frac{\log M}{\kappa} \le s_\kappa(\mathcal{S}) \le \min_\ell c_\ell. \tag{3}$$

For differentiable constituents, direct differentiation gives

$$\nabla s_\kappa(\mathcal{S}) = \sum_{\ell=1}^{M} w_\ell \nabla c_\ell, \qquad w_\ell = \frac{e^{-\kappa c_\ell}}{\sum_{r=1}^{M} e^{-\kappa c_r}}. \tag{4}$$

With overbars denoting division by fixed positive scales, the hierarchical certificate is

$$\begin{aligned} h_{\mathrm{rec}}(z) &= s_{\kappa_B}\left(\{\bar{\psi}_i(z)\}_i \cup \{\bar{h}^{\mathrm{b}}_i(z,\tau_k)\}_{i,k} \cup \{\bar{h}^{\mathrm{b}}_{B,j}(z,T)\}_j\right), \\ h_{\mathrm{input}}(z) &= s_{\kappa_U}\left(\{\bar{I}^+_d(z), \bar{I}^-_d(z)\}_{d=1}^m\right), \\ h_{\mathrm{aug}}(z) &= s_{\kappa_O}\left(\{h_{\mathrm{rec}}(z), h_{\mathrm{input}}(z)\}\right), \end{aligned} \tag{5}$$

where the command-admissibility constraints are defined as $I_d^+ = u_{\max,d} - u_d$ and $I_d^- = u_d - u_{\min,d}$, for $d = 1, \ldots, m$. The positive sharpness parameters and normalization scales are specified in Appendix C.4. Applying (3) at each level of (5) shows that $h_{\mathrm{aug}} \ge 0$ implies every encoded margin is nonnegative, while its gradient follows recursively from (4).

The continuous composite constraint is defined as

$$L_{\hat{f}} h_{\mathrm{aug}} + L_{\hat{g}} h_{\mathrm{aug}} \hat{u} + \alpha_{\mathrm{aug}} h_{\mathrm{aug}} = a(z)\hat{u} + b(z) \ge 0, \tag{6}$$

with a, b defined in Section 2. This single half-space need not equal the intersection of the separately enforced constituent CBF half-spaces.

B.3 Closed-Form Projection: KKT Derivation

Since the optimization problem defined in (1) only has one composite constraint, the closed-form solution can be derived through KKT conditions. Here, $\Omega_t = a(z_t)\hat{u}_t^{\mathrm{nom}} + b(z_t)$. With multiplier $\mu \ge 0$, the Lagrangian is

$$\mathcal{L}(\hat{u}, \mu) = \frac{1}{2}\|\hat{u} - \hat{u}_t^{\mathrm{nom}}\|_2^2 - \mu\big(a(z_t)\hat{u} + b(z_t)\big).$$

Stationarity gives $\hat{u} = \hat{u}_t^{\mathrm{nom}} + \mu a(z_t)^\top$. If $\|a(z_t)\|_2 > 0$, complementary slackness yields

$$\mu^\star = \frac{\max\{0, -\Omega_t\}}{\|a(z_t)\|_2^2},$$

which recovers (2). If $a(z_t) = 0$, the surrogate input has no local authority over the certificate: $b(z_t) \ge 0$ makes every $\hat{u}$ feasible, whereas $b(z_t) < 0$ is locally infeasible.

B.4 Continuous Safety Proof

The following argument uses the assumptions of Theorem 2.1 and applies throughout the continuous closed-loop solution's interval of existence. An all-time guarantee additionally requires forward completeness.

Proof of Theorem 2.1. Continuous enforcement of (6) gives $\dot{h}_{\mathrm{aug}} \ge -\alpha_{\mathrm{aug}} h_{\mathrm{aug}}$, hence $h_{\mathrm{aug}}(z(t)) \ge h_{\mathrm{aug}}(z(0))e^{-\alpha_{\mathrm{aug}} t} \ge 0$. The LSE bound and positive normalization therefore make every encoded margin nonnegative. In particular, $\psi_i = \dot{h}_i + k_0 h_i \ge 0$ implies $h_i(x(t)) \ge h_i(x(0))e^{-k_0 t} \ge 0$, while $I_d^\pm \ge 0$ ensures $u(t) \in \mathcal{U}$. The remaining margins enforce backup safety at each τ_k and terminal membership in $\mathcal{C}_B^z$. Backup invariance and $E_x\mathcal{C}_B^z \subseteq \mathcal{C}$ ensure safety when continuing the backup beyond T. □

C Quadruped Navigation Specialization

C.1 Policy Observation and Action

The actor observes normalized body-frame goal displacement, measured translational velocities and yaw rate, the previous executed command, normalized LiDAR ranges, the previous nominal command, and an indicator of its availability:

$$o_t = \begin{bmatrix} (\bar{p}^B_{G,t})^\top & \nu_t^\top & (u^{\text{safe}}_{t-1})^\top & \bar{\rho}_t^\top & (u^{\text{nom}}_{t-1})^\top & \iota_{t-1} \end{bmatrix}^\top.$$

Here, $\bar{p}^B_{G,t} = p^B_{G,t}/d_{\text{norm}}$, where $p^B_{G,t}$ is the body-frame goal displacement and $d_{\text{norm}} = 7.5\,\text{m}$, $\nu_t = [v^{\text{meas}}_{x,t}, v^{\text{meas}}_{y,t}, \omega^{\text{meas}}_{z,t}]^\top$ defines the measured velocities, and $\bar{\rho}_t$ contains ranges normalized by the maximum sensing range. The indicator $\iota_{t-1} \in \{0, 1\}$ controls the nominal-command finite difference in Appendix D. Let N_{scan} denote the number of LiDAR entries supplied to the actor, then the observation dimension is defined by $o_t \in \mathbb{R}^{N_{\text{scan}}+12}$.

The normalized action $\tilde{u}^{\text{nom}}_t \in [-1, 1]^3$ is mapped to physical commands by $u^{\text{nom}}_t = c_u + S_u \tilde{u}^{\text{nom}}_t$, where $c_u = (u_{\max} + u_{\min})/2$ and $S_u = \text{diag}((u_{\max} - u_{\min})/2)$. The bounds depend on the evaluation platform. The filter uses the current scan, incoming command state $u^-_t = u^{\text{safe}}_{t-1}$, and command history; goal displacement and measured velocity enter the actor.

C.2 Local Geometry and Backup Prediction

The reduced state is $x = [p_x, p_y, \theta]^\top$ and the applied command is $u = [v_x, v_y, \omega_z]^\top$, with dynamics

$$\dot{p} = R(\theta)[v_x, v_y]^\top, \qquad \dot{\theta} = \omega_z, \qquad \dot{u} = -Au + A\hat{u},$$

where $R(\theta)$ is the planar rotation matrix and $A = \text{diag}(a_x, a_y, a_\omega) \succ 0$. Each filter update initializes a local frame at $p = 0, \theta = 0$. A valid LiDAR return with measured range ρ_i and ray angle β_i defines the fixed point $q_i = p^B_L + \rho_i[\cos\beta_i, \sin\beta_i]^\top$, where p^B_L is the sensor offset in body coordinates. Its body-frame coordinate evolves as

$$\zeta_i(x) = R(\theta)^\top (q_i - p). \tag{7}$$

Safety margins. Let $r_x, r_y > 0$ be the longitudinal and lateral semi-axes of the inflated body ellipse, and define $\Pi = \text{diag}(r_x^{-2}, r_y^{-2})$. The barrier $h_i = \zeta_i^\top \Pi \zeta_i - 1$ requires each observed point to remain outside the inflated body ellipse. For $J = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, differentiating (7) gives $\dot{\zeta}_i = -[v_x, v_y]^\top - \omega_z J \zeta_i$, hence

$$\dot{h}_i = -2\zeta_i^\top \Pi [v_x, v_y]^\top - 2\omega_z \zeta_i^\top \Pi J \zeta_i.$$

The current-time constituent is $\psi_i = \dot{h}_i + k_0 h_i$. Its relative-degree properties and the backup-gradient chain rule are given in Appendix B.1.

Backup rollout. Under $\hat{u}_{\text{b}} = 0$, the command decays as $u(\tau) = e^{-A\tau} u_0$, and

$$\theta_{\text{b}}(\tau) = \theta_0 + \omega_{z,0} \frac{1 - e^{-a_\omega \tau}}{a_\omega},$$
$$p_{\text{b}}(\tau) = p_0 + \int_0^\tau R(\theta_{\text{b}}(\sigma)) \begin{bmatrix} v_{x,0} e^{-a_x \sigma} \\ v_{y,0} e^{-a_y \sigma} \end{bmatrix} d\sigma.$$

The implementation evaluates the planar integral using fixed eight-point Gauss–Legendre quadrature and differentiates that numerical map analytically. The certificate uses $K = 20$ positive-time samples $\tau_k = kT/K = 0.2k\,\text{s}$, with $T = 4\,\text{s}$, while the current-time constraint uses ψ_i. Future margins are defind through $h^{\text{b}}_i(z_0, \tau_k) = h_i(p_{\text{b}}(\tau_k), \theta_{\text{b}}(\tau_k))$.

C.3 Terminal Backup Set

Let $R_x = r_x + d_{\text{b}}$, $R_y = r_y + d_{\text{b}}$, and $S_B = \text{diag}(R_x, R_y)$, where $d_{\text{b}} > 0$ is the additional terminal clearance. For each observed point, define $\ell_i = \frac{1}{2} \log E_i$ with $E_i = \zeta_i^\top S_B^{-2} \zeta_i > 0$. Let

$c_\theta = \frac{1}{2}|\,(R_x/R_y) - (R_y/R_x)\,|$ and $\lambda_A = \min(a_x, a_y, a_\omega)$. Under $\hat{u}_\mathrm{b} = 0$, the residual-motion measure $V_\mathrm{b} = \left(v_x^2/R_x^2\right) + \left(v_y^2/R_y^2\right) + c_\theta^2\omega_z^2$ satisfies

$$\dot{V}_\mathrm{b} \le -2\lambda_A V_\mathrm{b}. \tag{8}$$

The terminal margins are $h_{B,i}^{\mathrm{geom}} = \ell_i$ and $h_{B,i}^{\mathrm{rec}} = \ell_i^2 - 2V_\mathrm{b}/\lambda_A^2$, giving

$$\mathcal{C}_B^z = \bigcap_i \{z : h_{B,i}^{\mathrm{geom}} \ge 0,\, h_{B,i}^{\mathrm{rec}} \ge 0\} = \bigcap_i \{z : \ell_i \ge \sqrt{2V_\mathrm{b}}/\lambda_A\}.$$

The terminal index j enumerates both margins for every valid ray.

Proposition C.1 (Terminal-set invariance)**.** *For the fixed-scan local model, $\mathcal{C}_B^z$ is forward invariant under $\hat{u}_\mathrm{b} = 0$ and satisfies $E_x(\mathcal{C}_B^z) \subseteq \mathcal{C}$.*

Proof. Set $w_i = S_B^{-1}\zeta_i$, $\bar{v}_B = S_B^{-1}[v_x, v_y]^\top$, and $M_B = S_B^{-1}JS_B$. Then $\ell_i = \log\|w_i\|_2$ and $\dot{w}_i = -\bar{v}_B - \omega_z M_B w_i$. Since the symmetric part of M_B has norm c_θ, on $\ell_i \ge 0$,

$$\dot{\ell}_i = -\frac{w_i^\top \bar{v}_B + \omega_z w_i^\top M_B w_i}{\|w_i\|_2^2} \ge -\|\bar{v}_B\|_2 - c_\theta|\omega_z| \ge -\sqrt{2V_\mathrm{b}}. \tag{9}$$

At the recoverability boundary, $\ell_i = \sqrt{2V_\mathrm{b}}/\lambda_A$. Equations (8) and (9) give

$$\dot{h}_{B,i}^{\mathrm{rec}} \ge -2\ell_i\sqrt{2V_\mathrm{b}} + \frac{4V_\mathrm{b}}{\lambda_A} = 0.$$

At $\ell_i = 0$ within the set, $V_\mathrm{b} = 0$, so translation vanishes and yaw is either zero or leaves the circular ellipse unchanged. Hence $\dot{\ell}_i = 0$, and the backup preserves the terminal set for all fixed scan points. Finally, $R_x > r_x$ and $R_y > r_y$ imply $E_x(\mathcal{C}_B^z) \subseteq \mathcal{C}$. □

C.4 LSE Normalization and Filter Parameters

We normalize current margins by $\bar{\psi}_i = \psi_i/s_\psi$, with $s_\psi = 1\,\mathrm{s}^{-1}$, and use unit scales for dimensionless future and terminal margins. Input margins use $\bar{I}_d^\pm = I_d^\pm/s_{I,d}$, where $s_{I,d} = \max(|u_{\min,d}|, |u_{\max,d}|)$ is computed from the platform-specific command bounds.

Simulation uses a 360° scan with 120 rays, $u_{\min} = [-1,-1,-1]^\top$, $u_{\max} = [2,1,1]^\top$, and $A = 8I_3\,\mathrm{s}^{-1}$. Real-robot evaluation uses a 180° scan with 61 rays, symmetric bounds $u_{\max} = -u_{\min} = [1, 0.6, 1]^\top$, and $A = 4I_3\,\mathrm{s}^{-1}$, where I_3 is the three-dimensional identity matrix.

D Finite-Rate Realization and Guarantee Scope

The filter is evaluated every $\Delta t = 0.1\,\mathrm{s}$. Let $u_t^- = u_{t-1}^{\mathrm{safe}}$ be the incoming command state, Σ a positive diagonal tracking gain, and $\iota_{t-1} \in \{0,1\}$ indicate whether the previous nominal command is available. The nominal surrogate is

$$\hat{u}_t^{\mathrm{nom}} = u_t^- + A^{-1}\left[\frac{\iota_{t-1}}{\Delta t}(u_t^{\mathrm{nom}} - u_{t-1}^{\mathrm{nom}}) + \Sigma(u_t^{\mathrm{nom}} - u_t^-)\right]. \tag{10}$$

When no previous command is available, the finite-difference term is set to zero. Applying (2) to (10) yields $\hat{u}_t^\star$. Exact integration of the command dynamics over one interval with this surrogate held constant gives

$$u_t^{\mathrm{pre}} = \Lambda u_t^- + (I_m - \Lambda)\hat{u}_t^\star, \qquad \Lambda = e^{-A\Delta t}, \tag{11}$$

where I_m is the m-dimensional identity matrix. The executed command is

$$u_t^{\mathrm{safe}} = \mathrm{clip}(u_t^{\mathrm{pre}}, u_{\min}, u_{\max}), \tag{12}$$

and becomes the incoming command state at the next update. Clipping in (12) enforces physical command bounds but does not guarantee satisfaction of the composite CBF condition between updates.

Theorem 2.1 applies to the continuous reduced model with fixed scan-defined obstacles. It does not certify intersample backup safety, finite-rate execution, full-body tracking errors, unobserved geometry, unmodeled obstacle motion, or arbitrary LiDAR noise.

Assumption D.1 (Experimental model applicability)**.** The reduced pose and command dynamics represent the high-level motion, with sufficiently accurate command tracking by the frozen $50\,\mathrm{Hz}$ locomotion controller.

E SAFTER–SAC Training Details

E.1 Critic and Actor Updates

The actor π_ϕ, parameterized by ϕ, produces a normalized nominal command $\tilde{u}_t^{\mathrm{nom}} \in [-1,1]^m$. Using the action center c_u and scaling matrix S_u defined in Appendix C.1, the executed action in critic coordinates is the safe action specified in Appendix D.

Replay samples $(o_t, \tilde{u}_t^{\mathrm{safe}}, r_t, o_{t+1}, \chi_t) \sim \mathcal{D}$ contain the executed action, reward r_t, and termination indicator χ_t, where $\mathcal{D}$ is the replay sampling distribution. True termination uses $\chi_t = 1$, while time-limit truncation uses $\chi_t = 0$ and stores the final observation before reset. For discount $\gamma \in [0,1)$ and entropy temperature $\alpha_{\mathrm{ent}} > 0$, the target is

$$\mathcal{Y}_t = r_t + \gamma(1-\chi_t)\left[\min_{i=1,2} Q_{\bar{\theta}_i}(o_{t+1}, \tilde{u}_{t+1}^{\mathrm{safe}}) - \alpha_{\mathrm{ent}} \log \pi_\phi(\tilde{u}_{t+1}^{\mathrm{nom}} \mid o_{t+1})\right].$$

Here, Q_{θ_i} and $Q_{\bar{\theta}_i}$ are critic i and its target network, with parameters θ_i and $\bar{\theta}_i$. The next nominal action is sampled from the actor and filtered to obtain $\tilde{u}_{t+1}^{\mathrm{safe}}$. Both critics minimize the mean squared error between $Q_{\theta_i}(o_t, \tilde{u}_t^{\mathrm{safe}})$ and the detached target $\mathcal{Y}_t$. With critic parameters fixed, the actor minimizes

$$J_\pi(\phi) = \mathbb{E}\left[\alpha_{\mathrm{ent}} \log \pi_\phi(\tilde{u}_t^{\mathrm{nom}} \mid o_t) - \min_{i=1,2} Q_{\theta_i}(o_t, \tilde{u}_t^{\mathrm{safe}})\right],$$

where the expectation is over replay observations and reparameterized actor samples. Critic gradients pass through the safety map, while entropy remains defined on the nominal policy.

Filter differentiation. Hold the filter state and command history fixed, and define $a_t = a(z_t)$, $b_t = b(z_t)$, and $\Omega_t = a_t \hat{u}_t^{\mathrm{nom}} + b_t$. For $a_t a_t^\top > 0$ and $\Omega_t \neq 0$, the local projection Jacobian is

$$J_{\mathrm{proj},t} := \frac{\partial \hat{u}_t^\star}{\partial \hat{u}_t^{\mathrm{nom}}} = \begin{cases} I_m, & \Omega_t > 0, \\ I_m - \dfrac{a_t^\top a_t}{a_t a_t^\top}, & \Omega_t < 0, \end{cases}$$

where I_m is the identity matrix. For the finite-rate map, the physical-action Jacobian is

$$J_{F,t} := \frac{\partial u_t^{\mathrm{safe}}}{\partial u_t^{\mathrm{nom}}} = D_{\mathrm{clip},t}(I_m - \Lambda) J_{\mathrm{proj},t} A^{-1}\left(\frac{\iota_{t-1}}{\Delta t} I_m + \Sigma\right).$$

The command gains A, Σ, interval Δt, history indicator ι_{t-1}, and propagation matrix $\Lambda = e^{-A\Delta t}$ are defined in Appendix D. Away from clipping thresholds, $D_{\mathrm{clip},t}$ is diagonal with entries one for interior components of u_t^{pre} and zero for saturated components. The normalized-action Jacobian is $S_u^{-1} J_{F,t} S_u$. The Identity ablation replaces only $J_{\mathrm{proj},t}$ by I_m during actor differentiation, preserving the forward filter map and all other derivatives.

E.2 Entropy Temperature and Critic Parameterization

Entropy regularization remains defined on the nominal stochastic policy, because its tanh-Gaussian log-density is tractable whereas the deterministic filtered action does not generally have a tractable induced density. The implementation uses automatic entropy tuning with $\theta_\alpha := \log \alpha_{\mathrm{ent}}$ and

$$J_\alpha(\theta_\alpha) = \mathbb{E}\left[-\theta_\alpha\left(\log \pi_\phi(\tilde{u}_t^{\mathrm{nom}} \mid o_t) + \mathcal{H}_{\mathrm{target}}\right)\right],$$

using $\mathcal{H}_{\mathrm{target}} = -3$ for the three-dimensional command. The log-probability is detached in this temperature update.

The critics are parameterized as $Q_\theta(o_t, \tilde{u}_t^{\mathrm{safe}})$. Because the correction reward depends on both the nominal and projected surrogate commands, distinct nominal actions can produce the same executed

action while incurring different correction costs. Moreover, nominal-command history is included in the subsequent observation. The critic can therefore alias such nominal-action choices and should be viewed as a practical observation-conditioned approximation rather than an exact Markov action-value representation. Accordingly, "exact" in the Jacobian discussion refers to the local analytical projection derivative, not to an exact policy gradient for the complete objective.

E.3 High-Level Reward Function

At each $10\,\mathrm{Hz}$ policy step, the reward is the weighted sum of the components in Table 6. All SAC-based methods in our framework use the same task terms, with the correction term disabled for Vanilla SAC and the correction-reward-off ablations.

Table 6: High-level reward components and signed weights. The correction weight is -0.05 by default and zero for Vanilla SAC and correction-reward-off ablations.

Component	Definition	Weight
Goal progress	$r_t^{\mathrm{prog}} = d_{t-1} - d_t$	$+4.00$
Goal reached	$r_t^{\mathrm{goal}} = \mathbb{I}[d_t \le d_{\mathrm{goal}}]$	$+30.00$
Time	$r_t^{\mathrm{time}} = 1$	-0.01
Timeout	$r_t^{\mathrm{timeout}} = \mathbb{I}[\mathcal{T}_t]$	-5.00
Forward alignment	$r_t^{\mathrm{align}} = \dfrac{\mathrm{ReLU}(v_{x,t}^{\mathrm{meas}})}{v_{x,\max}} \cos\beta_t$	$+0.20$
Reverse motion	$r_t^{\mathrm{rev}} = \left(\dfrac{\mathrm{ReLU}(-v_{x,t}^{\mathrm{meas}})}{v_{x,\max}}\right)^2$	-0.50
Yaw tracking	$r_t^{\mathrm{yaw}} = \left(\dfrac{\omega_{z,t}^{\mathrm{meas}} - \omega_t^{\mathrm{ref}}}{\omega_{z,\max}}\right)^2$	-0.05
Command smoothness	$r_t^{\mathrm{smooth}} = \lVert u_t^{\mathrm{safe}} - u_{t-1}^{\mathrm{safe}} \rVert_2^2$	-0.01
Filter correction	$r_t^{\mathrm{corr}} = \lVert (\hat{u}_t^{\mathrm{filt}} - \hat{u}_t^{\mathrm{nom}}) \oslash s_u \rVert_2^2$	$-0.05/0$
Unsafe termination	$r_t^{\mathrm{coll}} = \mathbb{I}[\mathcal{C}_t]$	-20.00

Here, d_t is the Euclidean distance to the goal, $d_{\mathrm{goal}} = 0.35\,\mathrm{m}$, and β_t is the signed body-frame goal bearing. The measured forward velocity and yaw rate are $v_{x,t}^{\mathrm{meas}}$ and $\omega_{z,t}^{\mathrm{meas}}$. Their command limits are $v_{x,\max} = 2\,\mathrm{m/s}$ and $\omega_{z,\max} = 1\,\mathrm{rad/s}$. The reference yaw rate is $\omega_t^{\mathrm{ref}} = \mathrm{clip}(k_\beta \beta_t, -\omega_{z,\max}, \omega_{z,\max})$, with $k_\beta = 1.5\,\mathrm{s}^{-1}$. The indicator $\mathbb{I}[\cdot]$ equals one when its condition holds, and $\mathrm{ReLU}(q) = \max(0, q)$. The event $\mathcal{C}_t$ denotes termination due to obstacle collision, boundary-wall contact, or falling. The event $\mathcal{T}_t$ denotes time-limit truncation without prior goal or unsafe termination. For SAFTER, we used $\hat{u}_t^{\mathrm{filt}} = \hat{u}_t^{\star}$, while for other filtered variants it denotes the corresponding filter output. The positive vector s_u contains componentwise correction-normalization scales, and $\oslash$ denotes elementwise division.

E.4 SAC Network Architecture and Optimization

All SAC-based methods in our framework share hidden layers of 512, 256, and 128 units with ELU activations. The actor is a reparameterized tanh-squashed diagonal Gaussian policy outputting three means and three log-standard deviations, the latter clamped to $[-5, 2]$. Twin Q-networks use separate Polyak-averaged targets with $\tau_{\mathrm{P}} = 0.005$. The actor, critics, and entropy temperature use Adam with learning rate 3×10^{-4}. Training uses discount $\gamma = 0.99$, target entropy -3, a replay capacity of 2×10^6 transitions, and minibatches of 1024. Updates begin after 20,000 environment transitions, with one update per environment step.